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Problem Chosen :	C

2024 APMCM summary sheet

This paper analyzes the development of **the pet industry** in China and globally.

For the first problem, Based on cats and dogs as the primary pet categories, data on pet food and medicine consumption in China over the past five years was collected. The data was normalized using dimensionless preprocessing, and a comprehensive evaluation was conducted by combining the **TOPSIS** entropy weight method with the Analytic Hierarchy Process (**AHP**). This approach provides an intuitive representation of the industry trends over the past five years. Influencing factors were identified through a literature review and further refined using the coefficient of variation method. An **ARIMA** time series model was constructed to forecast the development of China's pet industry over the next three years.

For the second problem, Data on the global pet industry, including the market sizes of pet food, healthcare, supplies, social entertainment, and smart devices, was collected and visualized. The analysis confirmed a consistent upward trend in the global pet industry, with pet food dominating the market. Market demand was analyzed based on the relationship between penetration rate and market size. A **weighted ensemble model** based on multi-source learning fusion was developed to predict global pet food sales revenue, the annual average pet food consumption per pet, and the global pet population over the next three years. Using the penetration rate formula, the global pet food market demand for the next three years was calculated.

For the third problem, based on the global pet food market demand trends and China's development status, this study, using the data in Attachment 3 as a foundation, supplements it with recent data on global pet food sales, as well as China's production, export volumes, and market revenue. After comparing multiple classical forecasting models, a **hybrid ARIMA time series and LSTM model** was selected. This model demonstrates high fitting accuracy, with correlation coefficients exceeding 80% and minimal error fluctuations, indicating robust performance. By solving this model, accurate predictions for China's pet food production and export over the next three years were achieved.

For the third problem, to account for the impact of new foreign economic policies from Europe and the U.S., tariffs and demand increments are quantified and integrated into the original export forecast model. The combined effects of tariffs directly influencing exports and indirectly suppressing global demand are modeled using a **Hybrid ARIMA-LSTM Tariff and Demand Model (HALT-D)**. This model predicts export volumes for the next three years, providing a basis for sustainable development strategies for China's pet food industry.

Keywords: the pet industry forecast ARIMA ensemble model LSTM

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I Introduction

1.1 Background

In recent years, the rapid development of the global economy, continuous improvement in per capita income, and significant shifts in consumer attitudes have driven the rise of the pet industry.

In China, the pet industry began in 1992, with the entry of international brands and the popularization of the "pet companionship" concept transforming pets from mere "animals" to "family members." This shift has fueled the rapid growth of related industries.

However, the development of the pet industry is influenced by various factors, such as increasing consumer attention to pet health, demographic and residential changes, the rise of e-commerce platforms, and the maturity of legal regulations and cultural awareness. Analyzing these factors, forecasting industry trends, and formulating effective development strategies have become critical research topics in the field.

1.2 Problem Statement

Based on the data provided in the appendices, including Appendix 1 (the number of pet cats and dogs in China over the past five years), Appendix 2 (the number of pet cats and dogs in the USA, France, and Germany), and Appendix 3 (China's pet food production and export volumes over the same period), this study aims to address the following questions:

Problem 1:

Analyze the development trends of China's pet industry over the past five years by pet type. Investigate the factors influencing this development, and construct a mathematical model to predict the development of China's pet market for the next three years.

Problem 2:

Analyze the development trends of the global pet industry and predict the global demand for pet food over the next three years.

Problem 3:

Based on the trend of global pet food market demand and the development of China, this paper analyzes the development of China's pet food industry, and establishes a mathematical model to forecast the production and export of pet food in China in the next three years without considering the change of economic policy.

Problem 4:

To develop new forecasting models taking into account the data in the annex, the additional data collected and the results of the above questions and the impact of European

and US foreign economic policies; Based on the forecast results, please formulate feasible strategies for the sustainable development of China's pet food industry.

II Model Assumptions

Assumption 1: The data obtained from searches is assumed to be reliable and reasonable.

Assumption 2: The primary pet categories are assumed to be cats and dogs, with the market share of exotic pets and other niche categories being negligible.

Assumption 3: It is assumed that the economic environment, policies (e.g., tariffs and subsidies), and technological advancements will remain stable over the next three years, with minimal impact on the industry.

Assumption 4: China's pet market trends are assumed to be relatively independent of those in other countries or regions, and international market data will not directly interfere with China's market predictions.

Assumption 5: The selected market samples are assumed to be representative and reflective of the overall trends in the global and Chinese pet industries.

III Notations

Symbol	Description
x_{ij}	The data of the i -th indicator in the j -th year.
p_{ij}	The proportion of the data for the i -th indicator in the j -th year.
e_j	The entropy value corresponding to the j -th year.
k	Standardization coefficient, used to ensure that the entropy value falls within the range $[0, 1]$.
d_j	The degree of data dispersion or variability.
w_j	The maximum eigenvalue reflects the importance of each year's data.
λ_{max}	The maximum eigenvalue

Y	The quantitative value of the development of China's pet industry
A	Total sales of pet food
B	Annual average expenditure per pet
C	Total number of pets
α	Food penetration rate

IV TOPSIS-AHP Analysis and ARIMA Prediction

4.1. Analysis and Solution

According to the requirements of the question, the development of China's pet industry in the past five years is analyzed from the perspective of pet types. Based on the data in Attachment 1, the paper firstly analyzes the types of pets in China by searching literature [1], which are mainly divided into dogs, cats and heteropets. In the analysis and research, it is found that the market of different pets is relatively small, so the development of China's pet industry in the past five years is mainly reflected by analyzing the market situation of dogs and cats. Based on the investigation of the existing literature [2] and the preliminary analysis of the factors affecting the development of China's pet industry: the consumption level of residents, the consumption level of rural residents, the consumption level of urban residents, the Engel coefficient of residents, population and so on. Due to the obvious correlation with these indicators and time, the data has a certain stationarity after analysis. Therefore, we consider establishing a goal planning model based on ARIMA time series forecast, and obtain a visual chart to reflect the development forecast of China's pet market in the next three years.

4.2.Data Collection and Processing

In order to more accurately reflect the situation of China's pet industry and reduce the difference in data sources, the White Paper of China's Pet Industry in relevant years was searched to obtain the market size of pet consumption of dogs and cats, the market size of pet food, the market size of pet drugs, the average annual consumption of each pet and the corresponding number.

4.1.1. An analysis of the development of China's pet industry in the past five years

In this study, some of the data can be directly visualized using time-series charts. This

approach provides a more intuitive and clearer representation of the data and its trends over the past five years, enabling an in-depth analysis of the development of China's pet industry, specifically focusing on dogs and cats.

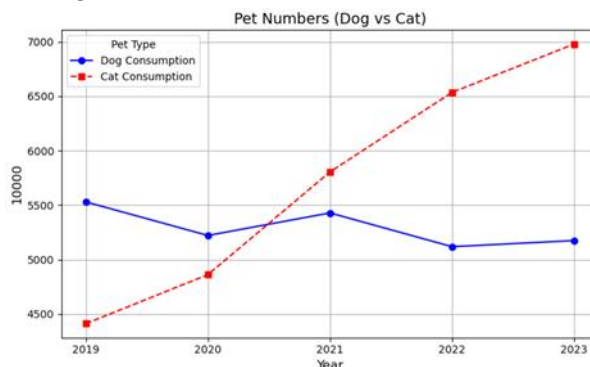


Figure1. Pet Population Change Chart from 2019 to 2023 (Cats, Dogs)

In 2023, the number of pet dogs will be 51.75 million, and the number of pet cats will be 69.8 million. For the fifth year in a row, the number of dogs was relatively flat, and the number of cats continued to rise.

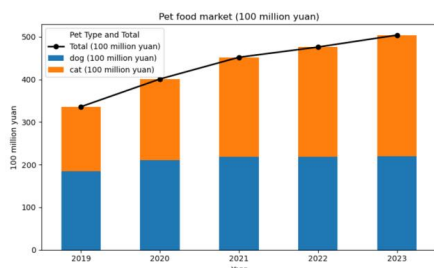


Figure2. Development and changes of pet food market from 2019 to 2023 (cats and dogs)

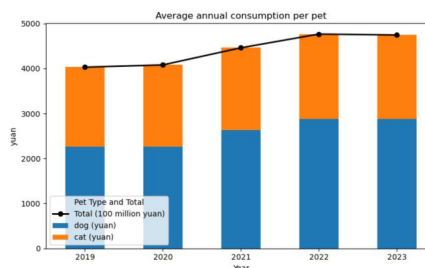


Figure3. Annual Average consumption change of individual pets (cats and dogs) from 2019 to 2023

According to Figure 2, the pet food market has continued to grow in the past five years, and the proportion of pet cats has increased year by year, while the growth of pet dogs is relatively slow. The pet cat food market has certain market potential. In Figure 3, the overall annual average consumption growth of a single pet.

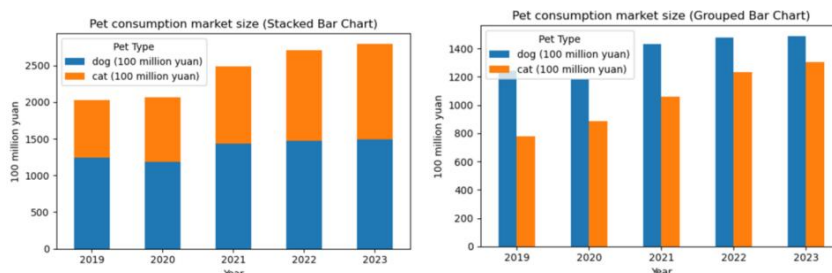


Figure4. 2019-2023 Development of China's pet industry in the past five years (Cat, dog)

The size of China's pet consumption market continues to grow, the size of the dog consumption market is growing slowly, and the size of the cat market continues to grow steadily.

In 4.1.1, the pet industry is subdivided into five parts. In order to obtain the Y value that can more accurately reflect the development of China's pet industry, the Y value needs to be calculated according to the proportion of each part to the total industry. Therefore, the

TOPIPS-AHP comprehensive analysis method is adopted to facilitate the statistics of Y value according to the statistical value. First, it needs to be dimensionless:

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^n x_{ij}}$$

The information entropy is then calculated using the following formula:

$$e_j = -k \sum_{i=1}^n p_{ij} \ln(p_{ij})$$

And for each year, the following formula is used to calculate the weight:

$$w_j = \frac{d_i}{\sum_{j=1}^m d_i}$$

The proportion of different markets in the pet industry is reflected according to the entropy weight method, as shown in Figure 5:

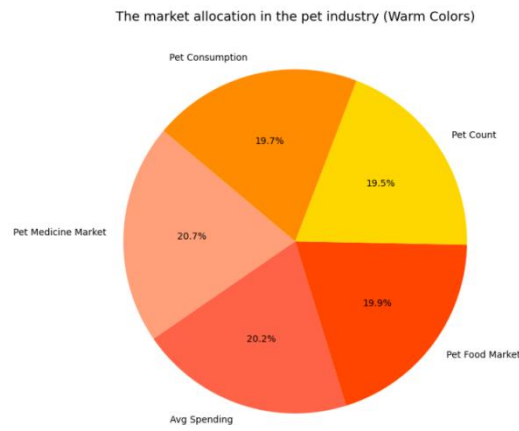


Figure5. China pet industry development market distribution

After obtaining the results of the distribution weights of the pet industry development market in China, the consistency test of the weights was carried out. If CR (consistency ratio) < 0.1, it can be judged that the matrix passes the consistency test and the weight is reasonable. Consistency check:

$$CI: CI = \frac{\lambda_{\max} - n}{n-1}$$

$$CR: CR = \frac{CI}{RI}$$

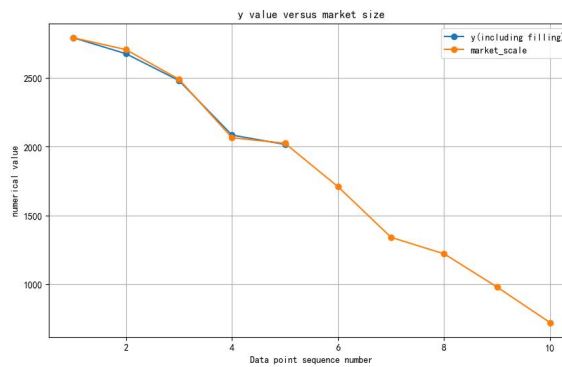


Figure6. Y-value and the changing trend of market size in China's pet industry

According to Figure 6, y value is highly consistent with the change trend of the market

size of China's pet industry obtained from the survey. Therefore, Y value is equivalent to the market size of China's pet industry.

4.1.2. Analyze the influencing factors

After determining the Y value, the influencing factors, such as residents' consumption level, residents' Engel coefficient, rural and urban population, residents' per capita disposable income and per capita GDP, were preliminarily determined according to the relevant literature survey. In addition, the data of these indicators in China in the past ten years is supplemented. Considering the relatively few data sets, the coefficient of variation method is used to analyze them, results are as follows:

Table1. Weight coefficient table of influencing factors

target	weight
Household consumption level	0.163791
Resident Engel coefficient	0.022220
Engel coefficient of urban residents	0.019989
Engel coefficient of rural residents	0.027925
Urban population	0.047454
Rural population	0.059672
Per capita disposable income	0.159046
Urbanization rate	0.048512
GDP per capita	0.0159505

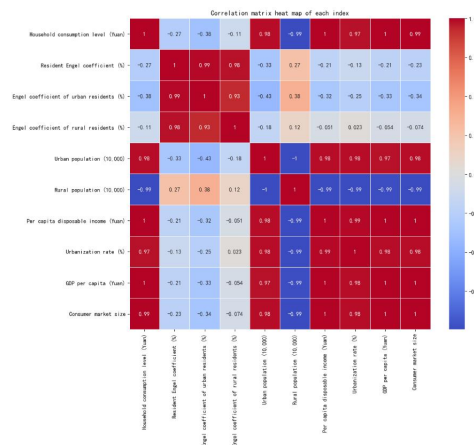


Figure7. Table of weight coefficients of each index

According to Figure 7, the consumption level of residents, the population of rural and urban areas, the per capita disposable income of residents and the per capita GDP have a great impact on Y value, indicating that promoting people's consumption can promote the growth of the pet industry.

4.3.ARIMA time series forecasting model [3]

By establishing ARIMA(p, d, q) model, the basic process of forecasting the pet development industry in China is carried out.

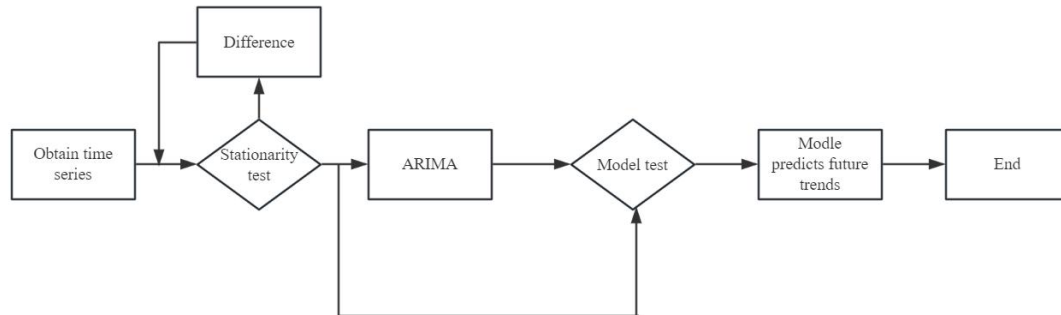


Figure8. Chinese pet industry development forecasting modeling process

Step1 stationary test (ADF test) :

In general, the formula of the P-order autoregressive process is defined as:

$$y_t = \mu + \sum_{i=1}^p \gamma_i y_{t-i} + \varepsilon_t$$

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_p X_{t-p} + u_t$$

The ADF test is performed on the original data, and the P-value is 0.9866. This p value is higher than the significance level (set at 0.05), indicating that the original sequence is non-stationary, and training the ARIMA model directly with the original sequence may lead to poor model results.

Step2 Sequence stabilization:

The original sequence is differentiated to make the mean of the time series no longer change with time, that is, to eliminate the trend and make the mean of the series stable. The difference is calculated as follows:

$$\Delta Y_t = Y_t - Y_{t-1}$$

Now check the original time series after first difference.

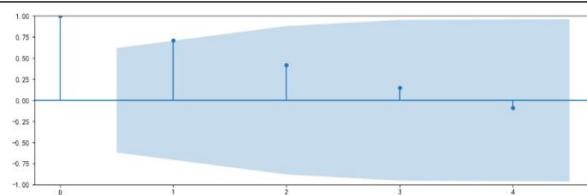


Figure9. autocorrelogram

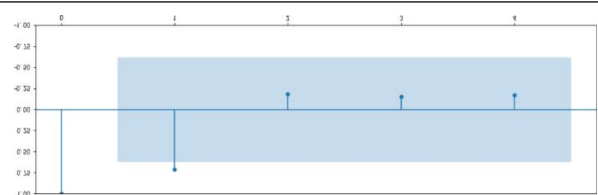


Figure10. Partial autocorrelation

It can be seen from Figures 9 and 10 that the autocorrelation coefficient of this time series presents a first-order trailing autocorrelation. From the partial autocorrelation figure 10, the partial autocorrelation coefficient shows a first-order truncated partial autocorrelation.

The result is shown below:

Table2. ADF test statistics

ADF test statistics	P
---------------------	---

difference preamplifier	0.5596	0.9866
backward difference	-6.7078	0.0000
Whether it is smooth or not	no	yes
Pass the test or not	no	yes

Step3 Model identification and order determination:

Identify the model according to the BIC criterion, and establish the ARIMA(2,1,0) model by fitting the values of p and q respectively through a cycle: ARIMA(p=2, d=1, q=0)

Step4 Prediction results:

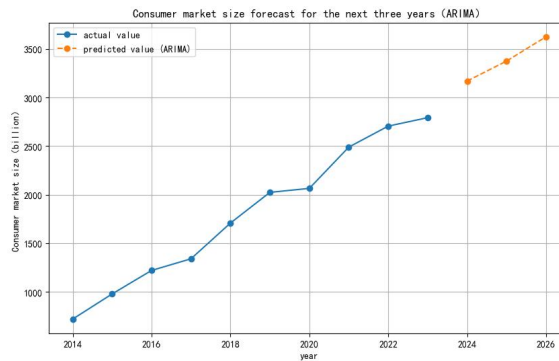


Figure11. Forecast result graph

Step5: Residual sequence analysis

The residual sequence extracted after model fitting is close to normal distribution, which proves that the current model fits the data reasonably without significant unexplained components, that is, the ARIMA model has a high reliability in predicting the future consumer market size.

4.4.Results:

Table3. China's pet industry development in the next three years (billion)

2024	2025	2026
3170.1981425	3375.901657666	3626.0750438

V A weighted ensemble model based on multi-source learning fusion

5.1. Development of the global pet industry

The global pet industry is subdivided into 5 major areas, and the relevant data is collected according to the Global Pet Industry White Paper over the years, so as to analyze the development of the global pet industry in recent years.

5.2. Major market analysis of the global pet industry

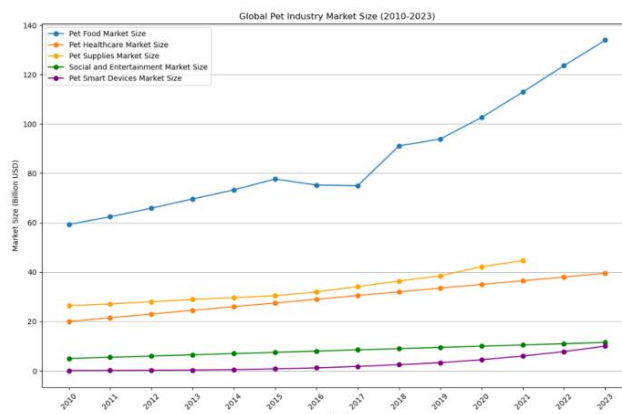


Figure12. Global pet industry market scale development

As can be seen from Figure 13, the global pet industry as a whole shows an upward trend. Even if the epidemic comes in 2020, it will not have a big impact on the pet industry, showing the strong rigid demand of the pet industry. In particular, the pet food market occupies the main part and grows rapidly after 2019, with great market potential. In contrast, the scale of the social and entertainment market grows very slowly, and the pet medical market and pet supplies market also have certain market potential.

5.3. Market share and growth rate analysis of pet industry in various countries around the world:

From the industry Research companies Euromonitor International, Statista, Grand View Research collected the TOP countries in the global pet industry year-on-year growth rate in the past three years, as well as the pet market share of various countries in the world.

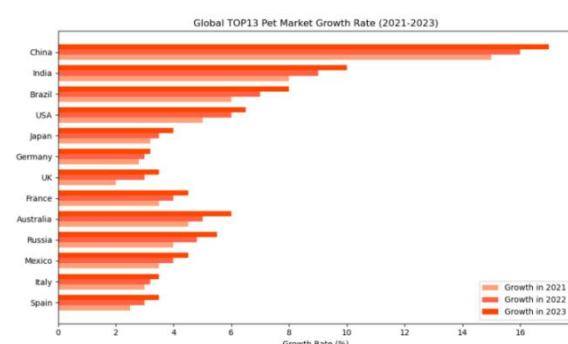


Figure13. the global pet industry growth rate is the TOP country

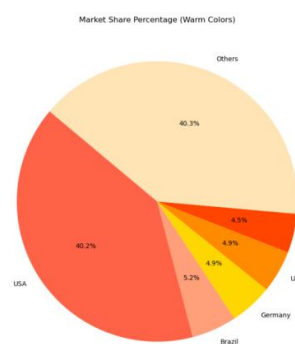


Figure14. Global pet market share by country

Fastest Growing markets: China and India continue to show strong growth, demonstrating the market potential of the Southeast Asian region.

The growth of mature markets: such as the United States, Germany, etc., although the growth rate is relatively slow, the market size is huge, and the demand is still strong.

Latin American market: The pet market in Brazil, Mexico and other Latin American countries is growing steadily, especially the demand for pet food and pet health products

continues to rise.

European markets: European countries such as France and Italy will see relatively stable growth in 2021, with demand focused on pet food and health care products.

5.1.3 Segmentation analysis of the global pet industry (food industry)

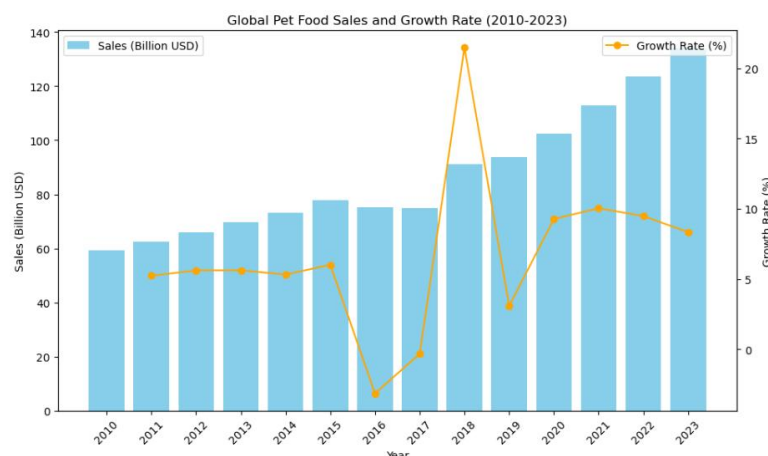


Figure15. Global pet food market and size

From 2010 to 2023, the global sales of pet food showed a steady growth trend, indicating that the pet market continues to expand and the demand for pet food is increasing. From 2020, although the growth rate has decreased slightly, sales still maintain rapid growth, indicating that the epidemic may accelerate the expansion of the global pet consumption market. In 2021-2023, the growth rate levelled off, probably because the market gradually matured and growth entered a stable phase, indicating that the pet food industry is highly competitive and the market size has expanded, but the new substantial growth potential may require more innovation.

5.1.4 Next, observe the distribution of pet food production in different regions of the world in the past two years:

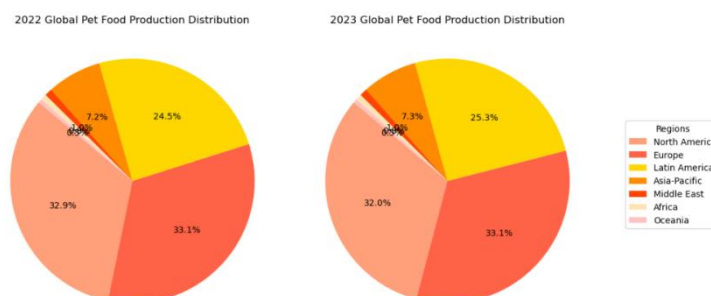


Figure16. Distribution of pet food production by region worldwide

North America and Europe accounted for the highest share, reflecting their developed pet food industry and mature market base, while the share of Asia Pacific increased slightly, reflecting the potential of the Asia Pacific market.

5.1.5 Pet food export volume in the past two years:

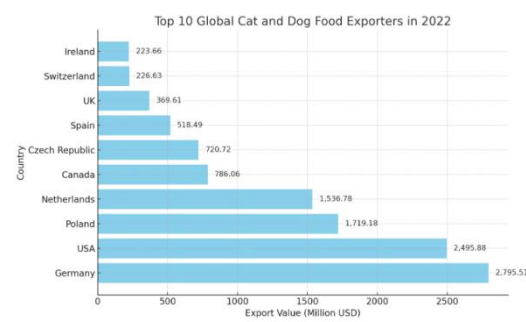


Figure17. TOP countries by food exports in 2022

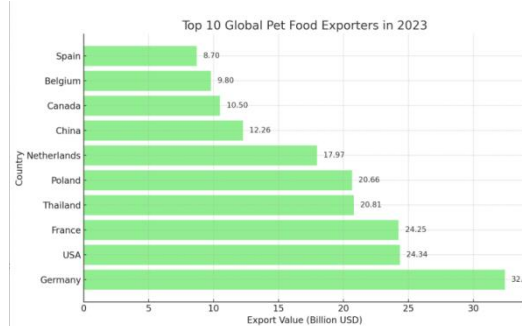


Figure18. TOP countries by food exports in 2023

Overall, compared with 2022, the international pet food industry trade market in 2023 is active, with Germany and the United States accounting for a relatively large share of the market, but the potential of the Asia-Pacific region is also gradually emerging.

5.1.6 Pet population analysis in some countries

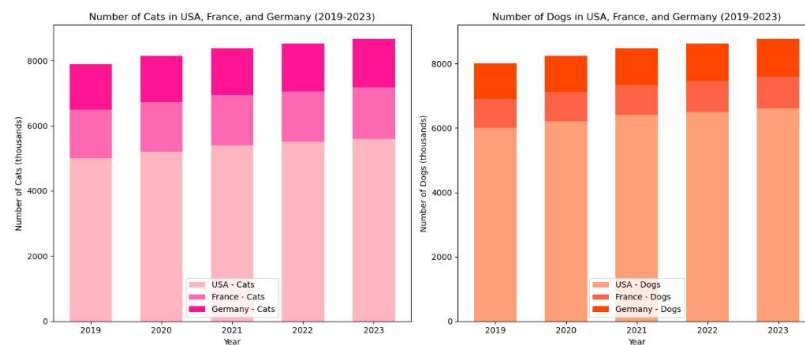


Figure19. The number of pets in Europe and America represented by the United States, France and Germany

The number of pet cats is relatively stable, the number of pet dogs is growing slowly, and the United States is more inclined to keep pets, and the number is far more than France and Germany. Products and services targeted at the dog market (such as dog food, training, health care) have greater growth potential, especially in the United States, and business strategies should focus on expansion opportunities in the dog market while meeting the needs of the mature cat market through innovation.

5.4. Weighted ensemble prediction model based on multi-source learning fusion

There is usually a positive correlation between penetration rate and market demand. Market demand can be predicted by analyzing the trend of penetration rate and market size. The penetration rate of the pet food market is calculated as follows:

$$\alpha = \frac{A}{C} \times \frac{1}{B}$$

In this paper, SVR model, linear regression model and ARIMA model are used to analyze and predict the original data, and their results are weighted to establish a weighted integrated prediction model based on multi-source learning fusion, so as to obtain the prediction results of global food penetration. The equations of linear regression and SVR are as follows:

Linear regression:

$$y_{actual} = (\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3) \cdot \sigma_y$$

SVR:

$$y_{actual} = \left(\sum_{i=1}^n (\alpha_i^* - \alpha_i) K(x_i, x) + b \right) \cdot c$$

The relevant parameters and results are predicted as follows:

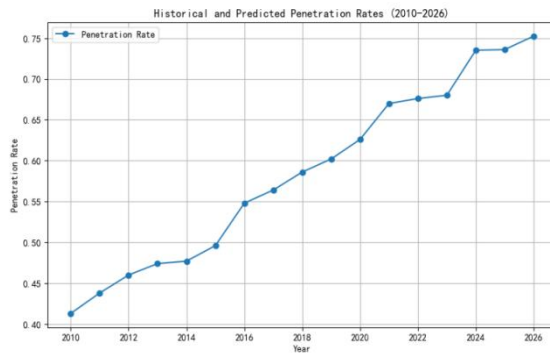


Figure20. Global pet food market penetration

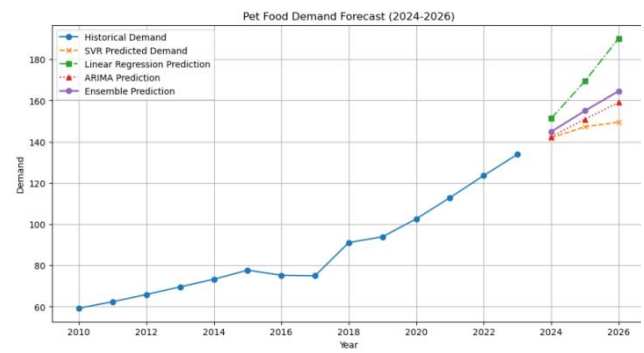


Figure21. Global pet food market demand forecast

accuracy test:

Table4. Test the accuracy of our model against other models

	SVR	ARIMA	Our model
MSE	31.1456	114.3442	58.1523
MAPE	2.58%	4.94%	3.53%

Combined with figure 20-23, our model forecast shows that demand will continue to maintain a steady growth trend, and the growth rate will slow down year by year. After accuracy test, the mean value of mape and mse of our model is low, and the single model is unstable in the face of different data trends. The integrated model can have stable performance in the face of different models, and the prediction performance is good. The prediction value of linear regression model is high, while the prediction value of SVR model is low. Our ensemble learning model combines the strengths of each model.

VI The development, production and export forecast of pet food industry in China

Combined with the current global pet market and the development of China, this paper analyzes the current industrial chain and pet food market of China's pet food industry and summarizes the development of China's pet food industry.

6.1 Development analysis of pet food industry in China:

6.1.1 China's pet industry chain

According to Wanlian Securities - pet industry segmentation track analysis, in China's pet industry chain, food and medical consumption account for the largest proportion. Pet food and medical coverage pets have a long life cycle, accounting for about 80% of the pet consumption market, of which China's pet food consumption has declined year by year, accounting for 50.7% in 2022, and pet medical consumption has risen steadily, accounting for 29.1% in 2022.

6.1.2 Pet Food Market in China:

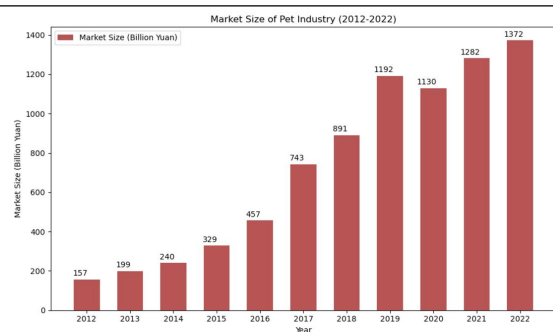


Figure22. Scale development of pet food in China

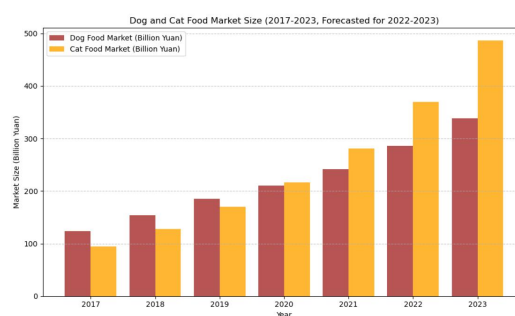


Figure23. China's dog food, cat food market scale
development

From 2012 to 2022, the size of China's pet food market has shown a year-on-year growth trend. Despite the impact of the epidemic, the size of pet food market has declined in 2020, and China is still one of the fastest growing regions in the global pet food industry. In recent years, with the growth of the number of pet dogs raised by Chinese residents and the upgrading of consumption concepts, the demand for pet dog food has continued to increase and entered a stage of rapid development. At the same time, with the rise of cat raising craze, more and more people will pet cats as a pet choice, and young pet owners are gradually increasing their attention to pet cats.

Pet staple foods occupy a dominant position in pet food. Compared with mature markets such as the United States and Europe, China's pet staple food market is in a period of rapid growth.

6.1.3 Domestic and international food brands

Global Pet Food Market Share (Warm Colors)

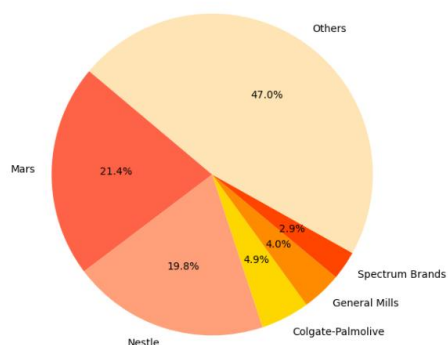


Figure24. The competitive landscape of global pet food enterprises in 2020

China Pet Food Market Share (Warm Colors)

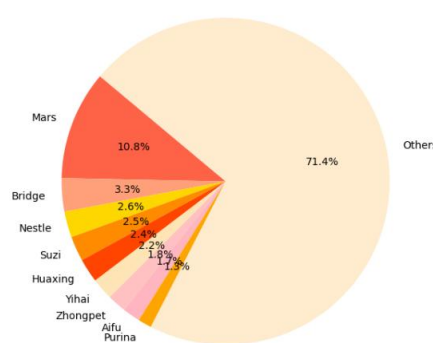


Figure25. Market share of pet food companies in China in 2020

The global pet food market is relatively concentrated, Mars, Nestle and other leading enterprises occupy half of the country. Compared with the international market, the domestic pet food market concentration is low. At present, the concentration of domestic pet food industry is low, and foreign brands occupy most of the market share. According to Euromonitor data, the industry concentration rate of the top five and top ten companies in 2020 is only 23.1% and 32.9%, and the first is foreign-owned Mars, with a market share of 10.5%. Due to the early start of European and American manufacturers, with scale cost advantages and years of accumulation of good brand image, the domestic pet food market is still dominated by European and American brands. Domestic pet food industry is still in the early stage of development, domestic enterprises are small, barriers have not yet fully formed, so the pet food competition pattern is more dispersed.

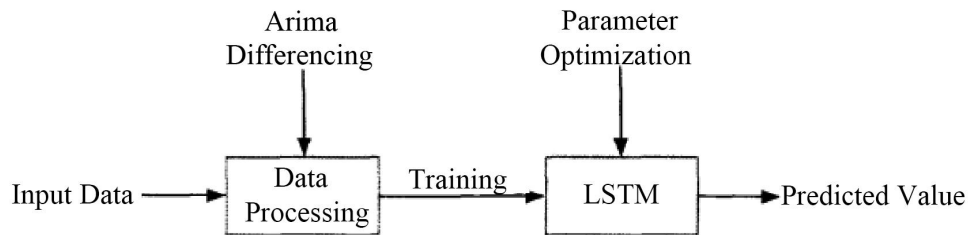
6.1.4 Conclusion

- Food and medical consumption account for the largest proportion in the pet industry chain. Among them, staple food consumption accounts for the largest proportion, and local enterprises are rising. It can be divided into pet staple food, pet snacks and pet health products, and pet staple food occupies a dominant position in pet food.

- China's pet food industry started late, and the compound annual growth rate of the market size is significantly higher than that of the United States, Japan and other developed countries. With the increasing number of pets in China and the increasing penetration rate of pet breeding, China's pet food market is expected to continue to grow.

6.2. Multiple models were established based on time series to optimize the results[4]

After the above analysis, the production and export of China's pet industry will be affected by global demand, domestic development, policy environment and other multiple factors, which involve many years of historical data, LSTM can use the past trend information to improve the accuracy of the forecast. At the same time, the pet food market has shown a rapid growth trend in recent years, and this dynamic change puts higher requirements on the traditional model, while the LSTM model has a strong dynamic adaptability, and can maintain a high prediction accuracy in the case of rapid data changes. Considering the characteristics of ARIMA model, the hybrid forecasting model formed by combining LSTM and ARIMA model can take into account short-term fluctuations and long-term trends. The LSTM model process is as follows:



The overall algorithm of ARIMA + LSTM can be represented by a primary formula as:

$$X_{t+n} = ARIMA(X_t) + LSTM(Residual_t)$$

among:

1. $ARIMA(X_t)$: Represents the long-term trend component extracted by the ARIMA model, used for forecasting long-term variations.:

$$Trend_{t+n} = ARIMA(X_t)$$

2. $Residual_t$: Represents the residual sequence of the time series, calculated as the actual value minus the trend value predicted by the ARIMA model.

$$Residual_t = X_t - Trend_t$$

3. $LSTM(Residual_t)$: Represents the nonlinear prediction of the residual sequence by the LSTM model, used to supplement the complex features that the ARIMA model fails to capture.

$$Residual_{t+n} = LSTM(Residual_{t-n}, \dots, Residual_{t+n})$$

4. predicted value (X_{t+n}): The final time series prediction value is obtained by adding the trend value predicted by ARIMA and the residual value predicted by LSTM:

$$X_{t+n} = Trend_{t+n} + Residual_{t+n}$$

To identify the optimal model, MAE, MSE, and MAPE were compared across predictions. MAE measures the average deviation, with smaller values indicating higher accuracy. MSE highlights the impact of large errors, reflecting overall stability, while smaller values indicate better performance. MAPE, expressed as a percentage, enables error comparison across different data scales, where lower values signify better predictions. Models tested include multiple linear regression, ARIMA, LSTM, and a combined ARIMA-LSTM model. The prediction results are as follows:

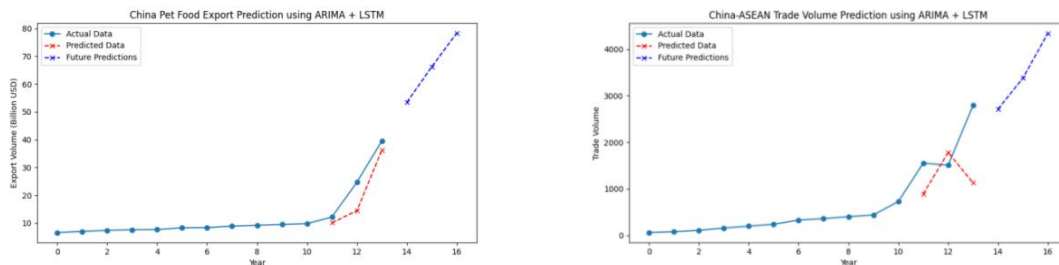


Figure26. Prediction results of ARIMA time series and LSTM combined prediction model

After the prediction results of the four models are obtained, MAE, MSE and MAPE values are calculated and compared. It was found that ARIMA forecast performed poorly in export prediction, especially MAPE up to 40.47%, indicating that the relative error was large and the model's fitting ability to small-scale export data was weak. LSTM forecast has the best fitting degree. However, due to the rapid growth trend of data this year, although the accuracy is the highest, LSTM does not consider the combination, resulting in too high forecast value and does not fit the reality. In the training process of the predictive model based on time series ARIMA, only the stage values corresponding to the fixed order can be learned, while the predictive model based on LSTM network has the ability to learn long-term dependent information. Therefore, the advantages of the two models are combined to achieve better prediction results. There is a certain training error, but it is more suitable for practice, so the model is finally selected, that is, the result in this case is the prediction result of the

ARIMA time series and LSTM combined prediction model mentioned above.

VII Hybrid ARIMA-LSTM Tariff and Demand Model (HALT-D)

7.1 Problem Analysis

7.1.1. Build a mathematical model

By establishing mathematical models, quantitatively analyze the specific impact of newly implemented foreign economic policies in Europe and the United States (such as tariff policies) on China's pet food industry. The core task is to construct a model capable of accurately capturing changes in export volume or sales, thereby evaluating the effects of tariff policies. The model will integrate historical data and collected exogenous variables, such as tariff data and global demand, to predict the potential impact of tariff changes on the industry.

7.1.2. Predicting Future Trends

Based on the constructed model, forecast the export volume or sales revenue of China's pet food industry in the coming years, particularly considering the potential impact of tariff policies. By comparing industry performance under scenarios with and without tariffs, evaluate the specific effects of tariff policies on industry development.

7.1.3. Providing Strategic Recommendations for Sustainable Development

Based on the model's predictions and analysis, formulate specific and feasible strategies to help China's pet food industry address the challenges posed by tariffs. These strategies include diversifying markets, adjusting production and supply chain strategies, and optimizing pricing strategies to mitigate the impact of tariffs. The core objective of these strategies is to reduce external policy risks, promote sustainable industry development, and ensure stable growth in export volume.

Through problem analysis, the core objective of this task is to establish an appropriate mathematical model to quantitatively analyze the impact of tariff policies on China's pet food industry, providing theoretical support and practical strategies for the industry's sustainable development in the future.

7.1.4 Introducing Exogenous Variables to Enhance Model Interpretability

Exogenous variables (Exogenous Variables, X) refer to variables that are not influenced by the time series data itself but may significantly impact it.

7.1.5 Tariffs: In international trade, tariffs directly affect the price and demand for exported goods. Changes in tariff levels typically lead to significant variations in export volume.

7.1.6 Global Demand: Fluctuations in global market demand also impact China's pet food exports. Changes in global demand are usually positively correlated with China's export volume; that is, an increase in global market demand tends to boost China's exports.

7.1.7 Advantages of the SARIMAX Model

In traditional ARIMA models, only past time series values are used to predict future values. However, a limitation of ARIMA is that it does not account for the influence of external factors or trends. To address this issue, we use the SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) model. By incorporating exogenous variables, the SARIMAX model can more accurately capture factors that influence the target variable.

7.1.8. Combining LSTM with Exogenous Variables

LSTM (Long Short-Term Memory network) is a deep learning model well-suited for processing sequential data, particularly excelling in time series forecasting. Its advantages become especially prominent when combined with exogenous variables:

- Sequence Memory Capability:** LSTM can retain long-term sequence dependencies, making it ideal for capturing long-cycle fluctuations or trends in time series data.

- Multivariable Handling:** By feeding residuals along with exogenous variables (e.g., tariffs) into the LSTM, the model not only learns the intrinsic structure of the time series (such as seasonality and trends) but also captures the impact of external factors.

- Improved Prediction Accuracy:** By considering exogenous variables, LSTM provides more accurate future output predictions. Particularly when policy-driven factors such as tariffs have a significant impact on exports, LSTM can adjust its forecasts dynamically and effectively.

7.1.9 Our Approach to Using Exogenous Variables

In this model, we integrate exogenous variables (e.g., tariffs) with time series data in the following ways:

- ARIMA + Exogenous Variables:** First, ARIMA is used to extract the long-term trend of the time series. Then, the residual sequence and tariffs are used as inputs, and LSTM is employed to predict the remaining components.

- LSTM Handling Exogenous Variables:** Historical residual data and tariffs (exogenous variables) are input into the LSTM model together. The LSTM learns how these external factors affect the time series, thereby improving the accuracy of the predictions.

7.2 The establishment of the new model

By incorporating exogenous variables (such as tariffs and global demand) into the model, it becomes capable of making more accurate and reasonable predictions when dealing with dynamic real-world scenarios. For industries like China's pet food sector, which are significantly influenced by policies and global market demand, introducing exogenous variables is crucial. This not only enhances the model's accuracy but also strengthens its interpretability, aiding in understanding future trends and making informed decisions.

Model Execution

After determining the modeling method, we performed calculations to predict the export volume for the next three years. The results are shown in the figure below:

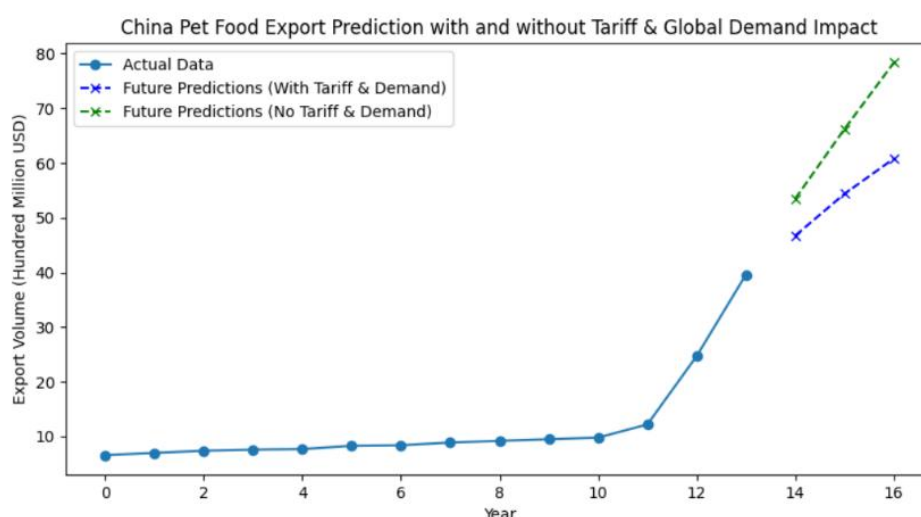


Figure 37: Forecast of China's Pet Food Export Volume for the Next Three Years

When tariff and global demand effects are excluded, the forecast shows a roughly stable growth trend in export volume. However, with the combined influence of tariffs and global demand incorporated, the export values are moderately adjusted. Increased tariffs lead to a reduction in export values, while growth in global demand partially offsets the tariff impact, resulting in a limited decline in export volume. The total export volume under the influence of tariffs is reduced by a certain amount compared to the scenario without tariffs.

Under the suppression of tariffs, export volumes for the next three years show a significant decline. As tariff increases reach 5%, 10%, and 15%, the reduction in export volume becomes progressively more pronounced. The relationship can be illustrated in the following chart:

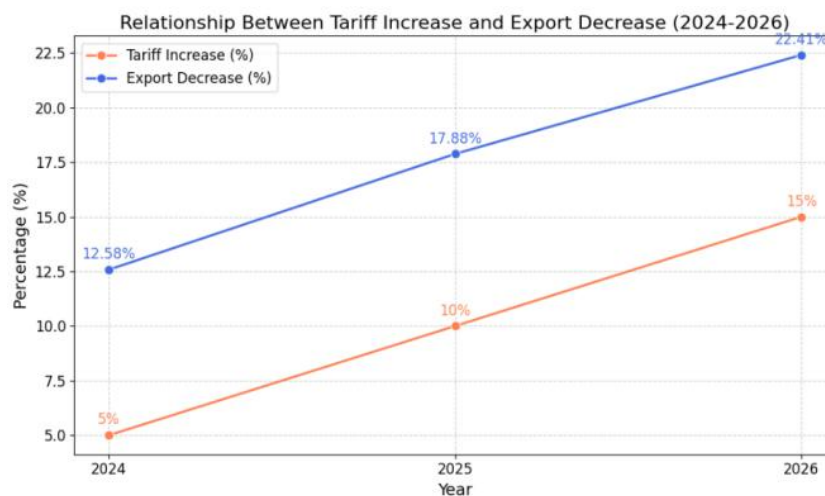


Figure 38: Relationship Between Tariff Increases and China's Pet Food Export Volume

As tariff levels progressively increase, the decline in exports exhibits a nonlinear intensifying trend, likely due to the gradual loss of price competitiveness.

7.3 Formulating Feasible Strategies for the Sustainable Development of China's Pet Food Industry

7.3.1. Enhancing International Market Competitiveness

Optimizing Product Quality and Brand Image

•**Objective:** Enhance international competitiveness by improving product quality and building brand influence, thereby mitigating the negative impact of tariffs on exports.

•**Measures:**

•Strengthen quality control in pet food production by adopting international standards (e.g., AAFCO, FEDIAF).

•Increase brand promotion efforts in target markets to establish a reliable and premium brand image.

•Launch customized pet food products to meet the needs of consumers in different countries (e.g., grain-free formulas, organic food).

7.3.2. Participating in International Trade Negotiations

- Objective:** Lower tariff barriers to ensure sustained industry growth.
- Measures:**
 - Strengthen collaboration with governments in target markets to reduce tariffs through bilateral and multilateral trade negotiations.
 - Advocate for signing free trade agreements (FTAs) with major export countries to secure tariff reductions for pet food exports.

7.3.3 Utilizing Preferential Policies

- Objective:** Fully leverage existing policy support to reduce production and export costs.
- Measures:**
 - Utilize export tax rebate policies to lower the tax burden on export enterprises.
 - Apply for national or local government incentive funds for high-quality export products.

7.3.4 Innovating Product Development

- Objective:** Meet the growing demands of global pet food consumption upgrades.
- Measures:**
 - Develop functional pet foods (e.g., for dental care, coat care, digestive health improvement).
 - Launch environmentally friendly pet foods (e.g., using sustainably sourced fish or plant-based protein alternatives).
 - Research formulas tailored for different pet species, ages, and health conditions.

7.4 Conclusion

By implementing the above strategies, China's pet food industry can more effectively address the challenges posed by tariffs and global demand fluctuations, enhance its global market competitiveness, and promote healthy and sustainable industry development.

VIII Evaluation and further generalization of the model

8.1 Advantages of the model

- In the first question, we combined methods such as TOPSIS, Analytic Hierarchy Process (AHP), Coefficient of Variation, and the ARIMA model, forming an analytical process that spans from industry composition to pet market forecasting.

- In the second question, we combined multiple models to leverage the strengths of each, enabling more accurate short-term predictions. This approach minimizes errors as much as possible and enhances flexibility.

• The ARIMA model excels at capturing linear trends and seasonal patterns in time series, effectively modeling the long-term trends of time series data.

• LSTM (Long Short-Term Memory) is a deep learning model capable of learning nonlinear relationships and complex short-term fluctuations in time series,

8.2 Deficiencies of the model

The collected data sample size can be further expanded, providing room for model optimization and improvement.

When the data is too sparse or insufficient, the model's predictions may be biased, so additional data collection is necessary.

The model combining ARIMA and LSTM is more complex compared to a single model, with a larger computational workload that requires more time for training and tuning. The training process of LSTM also takes longer, especially with large-scale datasets.

8.3 Generalization of the model

Adopting larger data sets can be used for house price forecasting

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